

Problem Chosen

**A**

2025

HiMCM  
Summary Sheet

Team Control Number

**16980**

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**Summary (Abstract).**

We address **HiMCM 2025 Problem A**, *Emergency Evacuation Sweeps*, by modeling responders, civilians, and hazards as a graph-based, discrete-time POMDP on the building's rooms, corridors, and exits. Our environment simulates fire and smoke spread, civilian awareness delays and self-evacuation, responder staging, search times proportional to room area, congestion effects, and exposure-limited responder safety, providing a realistic testbed for sweep strategies in offices, daycare facilities, and warehouses. On top of this simulator, we implement a hybrid solution: a **deterministic greedy sweep planner** based on VRP/MAPF-style risk scoring and time-expanded routing, and a **multi-agent PPO policy that consumes GAT embeddings, action masks, and shared rewards** to learn hazard-aware, redundancy-aware sweeps. In a baseline one-story office with two responders, our framework produces coordinated sweep orders and all-clear times that reduce backtracking and concentrate redundancy on high-risk rooms rather than corridors. In extended **multi-floor daycare and warehouse layouts**, we use the same framework to identify near-optimal responder counts and starting locations while quantifying trade-offs between sweep time, survival, and high-risk redundancy under different hazard and occupancy patterns. Across scenarios, the greedy planner offers fast, interpretable sweep plans and staffing guidance, whereas the **graph-RL policies** adapt to stochastic hazards and civilian behavior to further improve safety metrics; together, they outperform **distance-only heuristics** and support actionable recommendations on responder staffing, fire-door placement, and sensor-driven building automation.

**Keywords:** Emergency evacuation sweeps; graph-based POMDP; multi-agent reinforcement learning; PPO; graph attention network (GAT); greedy sweep planner; VRP/MAPF; building fire safety.

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# 1 Introduction and Problem Restatement

During emergencies such as fires or gas leaks, trained responders must sweep buildings to ensure that all occupants have either evacuated or are being guided toward exits. The Clearing Of Multi-use Assembly Public/Private spaces (COMAP) has requested a mathematical model that optimizes these *sweeps* in multi-floor buildings, balancing three competing objectives: (i) clearing all rooms in the shortest possible time, (ii) preserving occupant safety, and (iii) preserving responder safety and workload.

We formulate emergency sweeps as a discrete-time process on a building floor plan represented as a graph: nodes are rooms, hallways, stairwells, or exits; edges are traversable connections. Occupants move stochastically toward exits once they become aware of the emergency, while responders follow sweep strategies that determine which rooms to enter, in what order, and how many times. Our model incorporates (1) heterogeneous rooms and occupancies (e.g., daycare vs. office vs. warehouse), (2) time-varying hazards (fire intensity and smoke), and (3) health-point (HP) dynamics for both occupants and responders, including differential vulnerability of children, adults, and limited-mobility individuals. A room can require more than one sweep (redundancy) before it is declared all-clear, reflecting real-world practice.

The 2025 HiMCM Problem A statement structures the work into five Tasks. For clarity, we restate these Tasks and indicate where each is addressed in our paper:

- **Task 1 (Understand the problem).** Identify the key elements that inform sweep strategies, such as criteria for declaring rooms all-clear, occupant types, number and expertise of responders, redundancy, and type of emergency. We address this in Section 2, which formalizes our modeling assumptions for emergency sweeps.
- **Task 2 (Create your building sweep model).** Build a model for a one-story office with two exits and two responders, and estimate the time to complete a sweep. In Section 3, we introduce our graph-based building sweep model, movement rules, HP and hazard dynamics, and the baseline sweep strategies used for the basic scenario.
- **Apply your model.** Apply the model to additional layouts with different floor plans, numbers of floors, occupancies, and responders, and compute optimal sweep assignments and completion times. Section 4 extends the model to daycare and warehouse layouts, and compares sweep order, responder starting positions, and sweep times across scenarios.
- **Task 4 (Expand and reflect).** Incorporate more realistic constraints such as time-varying hazards, limited occupant awareness, moving crowds, and fewer responders, and discuss how technology changes the picture. Section 5 extends our model with dynamic fire intensity and smoke fields, HP loss for both occupants and responders, and technology-assisted sensing, and evaluates robustness under these constraints.
- **Task 5 (Share your model and results).** Provide practical recommendations for evacuation sweep strategies tailored to different building types. Section 6 presents a one- to two-page letter to the Local Emergency Planning Committee summarizing our recommended sweep strategies, redundancy policies, and technology investments.

This structure keeps the paper aligned with the HiMCM rubric while allowing us to use a single, coherent simulation framework to compare classical rule-based sweep heuristics with data-driven strategies trained by reinforcement learning.

## Hybrid Sweep Multi-agent Simulation Environment

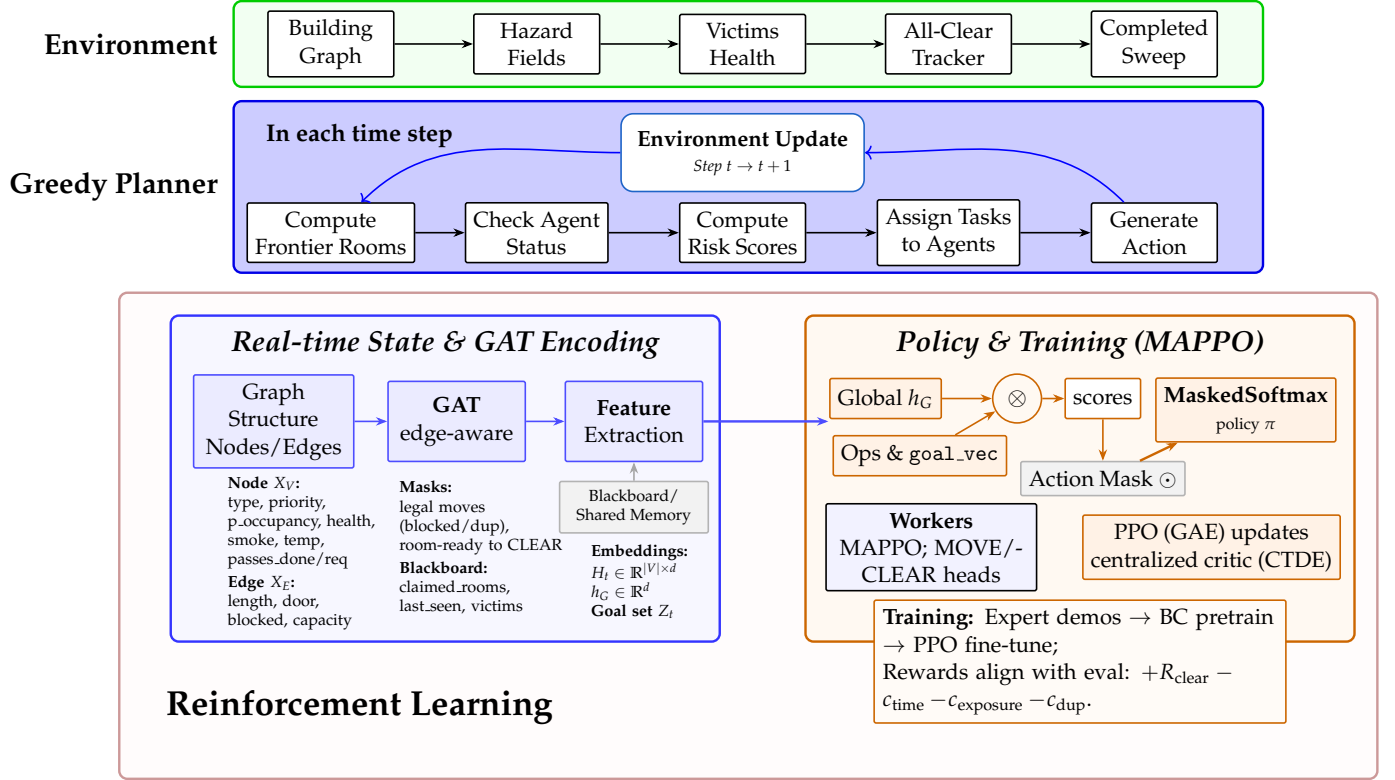


Figure 1: Overall sweep framework with explicit *goal\_vec*, action masking, shared rollout buffer, and a deterministic planner baseline. A graph-attention encoder produces node embeddings, and a shared multi-agent PPO policy (MAPPO under CTDE) selects feasible actions from masked logits; reward

## 2 Assumptions and Key Elements of Emergency Sweeps

To address Task 1, we specify the key elements that drive emergency sweeps and state the assumptions that make our model both tractable and realistic. These assumptions apply across all building layouts (office, daycare, warehouse) and are reflected consistently in our simulations.

### 2.1 Criteria for Declaring Rooms All-Clear

We assume that responders only declare a room “all-clear” after a thorough search consistent with real-world sweep practice. More precisely:

- Each room is assigned a required number of sweeps, denoted  $N_{\text{req}}(r) \in \{1, 2\}$ . Low-risk rooms (e.g., standard offices) typically have  $N_{\text{req}}(r) = 1$ , while high-risk rooms (e.g., daycare rooms, interior storage with poor visibility) have  $N_{\text{req}}(r) = 2$ .
- A *sweep pass* occurs when a responder spends the prescribed search time in the room, systematically checking all areas (e.g., behind furniture, under desks). We treat this as a discrete “search” action with duration proportional to the floor area of the room.
- A room is considered all-clear if and only if:
  1. it has been swept at least  $N_{\text{req}}(r)$  times, and
  2. no occupants remain in the room or on its incident edges (i.e., all occupants found there have either exited the building or are currently being assisted by a responder), and

3. hazard levels (fire intensity and smoke) remain below a critical threshold for responders to safely re-enter if needed.

This definition makes redundancy explicit: high-risk rooms are intentionally over-swept, while low-risk rooms are cleared after a single verified pass. It also distinguishes between *automatic evacuation* (occupants leaving on their own) and *formal all-clear*, which requires an intentional search.

## 2.2 Occupant Types and Room/Layout Heterogeneity

Occupants differ in both mobility and vulnerability. We explicitly model three categories:

- **Children** (e.g., daycare rooms): lowest walking speed and highest vulnerability to hazards.
- **Adults**: baseline walking speed and intermediate vulnerability.
- **Limited-mobility occupants** (e.g., elderly or disabled individuals): slower than adults and more vulnerable than both children and adults.

Each occupant starts with 100 health points (HP). At each time step, hazards in their current location reduce HP according to:

- a *fire intensity* level in  $[0, 1]$  (how strongly the room is burning),
- a *smoke density* level in  $[0, 1]$  (how obscured and polluted the air is), and
- a category-specific multiplier (children and limited-mobility individuals lose HP faster than healthy adults under the same conditions).

We define an occupant as *tenable* if their HP is above a category-dependent threshold. Tenable occupants can self-evacuate along shortest paths to the exits once they become aware of the emergency. When HP falls below this threshold, the occupant is *non-tenable* and can no longer self-evacuate; such individuals require physical assistance from a responder to reach the exit. This provides a clear, health-based definition of who “needs assistance” rather than treating assistance as an arbitrary flag.

Room layouts and occupancies vary by scenario:

- **Office**: small rooms with mostly adult occupants, moderate density.
- **Daycare**: moderate-sized rooms with a high proportion of children, higher vulnerability and stricter redundancy requirements.
- **Warehouse**: large, sparse rooms with mixed occupants and longer travel distances.

Table 1: Scenario matrix outlining occupancy heterogeneity and policy parameters.

| Scenario  | Typical room area (m <sup>2</sup> ) | # required passes | Occupant mix                                                                                |
|-----------|-------------------------------------|-------------------|---------------------------------------------------------------------------------------------|
| Daycare   | 20–80 (classrooms, playrooms)       | 2                 | Children 85–90%, Adults (staff) 10–15%; higher vulnerability; TIC recommended.              |
| Office    | 10–40 (offices/-classrooms)         | 1                 | Adults 85–95%, Limited mobility 5–10%; partitions may reduce visibility.                    |
| Warehouse | 80–400+ (open bays, racks)          | 2 (labs)          | Adults 90–95%, Limited mobility 5–10%; potential hazardous materials; fire doors impactful. |

These heterogeneities affect both sweep time (via room area and connectivity) and risk (via occupant mix and hazard evolution).

## 2.3 Number and Expertise of Responders

Responders are modeled as trained agents with symmetric capabilities unless otherwise stated. Each responder has:

- a fixed walking speed along hallways and through doors, calibrated so that traversing a typical corridor segment takes several time steps;
- a search rate, which determines how long it takes to sweep a room of given area;
- access to basic situational awareness (e.g., radio communications and a schematic floor plan) but no perfect foresight about where occupants are located;
- a health state, also measured in HP, that decreases under high fire intensity and smoke, representing heat stress, limited visibility, and toxic inhalation[1].

Responders are removed from the sweep if their HP falls to zero, representing incapacitation. This couples responder safety directly to sweeps: paths that repeatedly traverse high-intensity fire zones may clear rooms faster but at the cost of higher responder HP loss.

In most experiments we use the same responder capabilities across scenarios to isolate the effect of building layout and occupancy. When we vary the number of responders, we keep individual capabilities fixed and analyze how additional staff changes sweep time, occupants rescued, and responder HP loss.

## 2.4 Hazard Types and Their Representation

We focus on two hazard fields that evolve over time within the building:

- **Fire intensity** in each room or hallway, modeled as a continuous variable in  $[0, 1]$  that increases due to local burning and may spread to adjacent nodes.
- **Smoke density**, also in  $[0, 1]$ , which may spread more quickly than fire and affects both occupant HP loss and effective visibility.

Both hazards influence HP loss for occupants and responders. Fire intensity dominates HP loss at high levels (thermal injury), while smoke density dominates at lower levels (inhalation and confusion). We assume that responders can observe approximate fire and smoke levels throughout the building (e.g., via thermal cameras and smoke sensors[2]), but occupants only experience them locally.

While our baseline simulations focus on fires, the same framework can be adapted to other emergencies:

- **Gas leak**: treat smoke density as a toxic gas concentration field with higher HP penalties and reduced visual cues.
- **Earthquake**: model localized blockages and temporarily inaccessible rooms while reusing the same sweep and redundancy logic.

These extensions are discussed qualitatively in Section 5.

## 2.5 Redundancy Policy: Primary and Secondary Searches

Redundancy is built into our model through explicit required sweep counts per room:

- **Primary search**: the first sweep pass that verifies whether a room contains occupants and, if so, initiates evacuation or assistance.
- **Secondary search**: an additional sweep pass (when  $N_{\text{req}}(r) = 2$ ) used for rooms where the probability of missing someone is higher (e.g., complex geometry, high occupant density, or vulnerable populations).

Operationally, our redundancy assumptions are:

- Office rooms and simple hallways generally have  $N_{\text{req}}(r) = 1$ , unless they are interior spaces with no direct external windows.
- Daycare rooms and other high-risk spaces (e.g., lab rooms, storage areas with low visibility) have  $N_{\text{req}}(r) = 2$ , enforcing a secondary search before being declared all-clear.
- Exits and stairwells must be checked at least once but do not require a secondary sweep, as traffic flow provides implicit verification.

By encoding redundancy directly into  $N_{\text{req}}(r)$ , we can systematically compare sweep strategies that trade off:

- faster coverage with minimal redundancy versus
- more conservative policies that invest time in secondary searches of high-risk rooms.

This fulfills the Task 1 requirement to specify “amount and type of redundancy” in a way that is mathematically precise and directly usable in our subsequent model (Task 2) and simulations (Task 3).

### 3 Core Deterministic Sweep Model

In this section we construct a deterministic sweep model that captures the essential elements of emergency searches in the one-story office baseline. Responders follow simple, rule-based behavior: they traverse the building graph, search rooms with time proportional to area, assist civilians whose health or mobility requires help, and declare the building “all clear” once every search-relevant space has received the required number of passes. This model already answers Problem A and serves as the reference for our later reinforcement learning (RL) experiments.

#### 3.1 Graph Representation of Buildings

We represent the building as a graph

$$G = (V, E),$$

where each node  $v \in V$  corresponds to a physical location (room, hall segment, exit, or special space such as a nursery or playroom) and each edge  $(u, v) \in E$  is a traversable connection (door, corridor segment, stair).

Each node  $v$  is associated with geometric and semantic attributes:

$$\text{area}(v), \text{length}(v), \text{floor}(v), \text{ntype}(v), \text{required\_passes}(v),$$

encoding its floor area, characteristic path length, floor index, room type (e.g., office, hallway, nursery), and the number of times it must be swept in high-redundancy scenarios. These fields are implemented in the simulation by a `NodeMeta` structure.

Edges  $(u, v)$  carry attributes

$$\text{length}(u, v), \text{width}(u, v), \text{slope}(u, v), \text{door}(u, v), \text{fire\_door}(u, v),$$

stored in `EdgeMeta`. The `fire_door` flag will later reduce fire spread across that connection.

For the one-story office baseline,  $G$  takes the form shown in office graph Figure (three rooms on each side of a central corridor, with exits at both ends), which matches the description in the problem

statement.

### 3.2 Agents, Civilians, and Hazards

**Civilian mobility and awareness.** Each civilian  $p$  has a mobility type

$$\text{mobility}_p \in \{\text{child}, \text{adult}, \text{limited}\},$$

with base free-walking speeds

$$v_{\text{child}} = 0.9 \text{ m/s}, \quad v_{\text{adult}} = 1.2 \text{ m/s}, \quad v_{\text{limited}} = 0.7 \text{ m/s},$$

chosen slightly below the typical 1.3–1.4 m/s adult walking speed to reflect slower movement under stress and for vulnerable groups.

Each occupant is initialized with an awareness timer

$$T_{\text{aware}} \sim \mathcal{D}(\mu_{\text{aware}}, \sigma_{\text{aware}}),$$

representing the delay between alarm onset and the moment they decide to evacuate. Values in the range 50–220 s are consistent with behavioral studies and are implemented in our code as integer timesteps of length  $\Delta t = 5$  s.

Before the timer expires, civilians remain in an **UNAWARE** state; once it hits zero, they transition to an **EVAC** state and follow a shortest-path route to an exit (Section 3.3). HP degradation applies in both UNAWARE and EVAC when the occupant is inside fire or smoke.

**Responder staging, movement, and search.** We model  $k$  responders (agents), all spawned at designated entry nodes that coincide with exits in the graph. To account for turnout and size-up, each agent experiences a staging delay

$$t_{\text{staging}} = \text{staging\_delay} \approx 10 \text{ timesteps} \approx 50 \text{ s},$$

during which they cannot move, search, or assist.

Once active, movement along an edge  $e = (u, v)$  of length  $\ell_e$  is modeled as a time-consuming traversal at speed `agent_speed`:

$$t_{\text{move}}(e) = \max\left(1, \left\lceil \frac{\ell_e}{\text{agent\_speed}} \right\rceil\right) \text{ timesteps.}$$

During those  $t_{\text{move}}(e)$  steps the responder is marked as `on_edge = True` and cannot initiate new actions. If the responder is assisting a civilian, that person is moved along the same edge.

Searching a room is an exclusive action with time proportional to the room area. For a node  $v$  with floor area  $\text{area}(v)$  we use

$$t_{\text{search}}(v) = \max\left(1, \lfloor \gamma_{\text{search}} \cdot \text{area}(v) \rfloor\right), \quad \gamma_{\text{search}} \approx 0.5 \text{ steps/m}^2, \quad (1)$$

so that a 16 m<sup>2</sup> bedroom requires  $t_{\text{search}}(v) = 8$  steps ( $\approx 40$  s)—consistent with practical guidance for primary searches by small teams. While `searching = True`, the agent cannot move or start assistance. When the search completes, all civilians in  $v$  are marked as discovered, the aggregate node-level observations are updated, and the node's sweep counter is incremented (see Section 3.3).

**Assistance and tenability.** Civilians maintain a health variable  $h_p(t) \in [0, 100]$  and a tenability threshold  $h_p^{\min}$ . When  $h_p(t) > h_p^{\min}$  they may still self-evacuate along their route; once  $h_p(t) \leq h_p^{\min}$  they can survive but cannot reach an exit without assistance.

Responders can issue an `assist_{person_id}` action for a discovered civilian at their current node. The civilian then becomes attached to that responder and moves at an assisted speed

$$v_{\text{assisted}} = \kappa v_{\text{base}}, \quad \kappa \approx 1.8,$$

where  $v_{\text{base}}$  is their mobility-dependent speed ( $v_{\text{child}}, v_{\text{adult}}, v_{\text{limited}}$ ). Assisted civilians follow the responder's path deterministically instead of re-planning on their own. Each responder may assist at most one civilian at a time.

To approximate responder safety, each agent also accumulates a hazard exposure score; when this exceeds a configurable threshold  $\Xi_{\text{max}}$ , the agent is restricted from initiating new searches and must prioritize withdrawal. In the deterministic baseline we do not model responder death, but we do penalize scenarios that drive exposure close to or above  $\Xi_{\text{max}}$ .

**Fire, smoke, and HP decay.** Hazards are modeled at the node level by binary fields

$$\text{on\_fire}(v, t), \quad \text{smoky}(v, t) \in \{0, 1\}.$$

Fire spread is a discrete contact process: every  $d_{\text{spread}} = \text{fire\_spread\_delay}$  steps, each burning node attempts to ignite each neighbor via edge  $e$  with probability

$$p_{\text{spread}}(e) = \begin{cases} p_0, & \text{if } e \text{ is a regular door or opening,} \\ 0.3 p_0, & \text{if } e \text{ is a fire-rated door,} \end{cases} \quad (2)$$

where  $p_0 = \text{fire\_spread\_prob}$  is a global base spread probability. In all reported experiments we fix

$$p_0 = 0.30, \quad d_{\text{spread}} = 3,$$

so that an isolated room adjacent to a burning neighbor has a probability  $1 - (1 - p_0)^3 \approx 0.66$  of igniting over 15 s, which produces plausible corridor-level growth without making spread almost deterministic. Smoke deterministically propagates to all neighbors of burning nodes, so fire-rated doors reduce fire spread probability but not smoke transmission.

Health degradation per civilian is modeled nonlinearly to reflect faster deterioration at low remaining health. Let  $h_p(t) \in [0, 100]$  denote the HP of civilian  $p$  at time  $t$  and  $\Delta t = 5$  s the timestep. We define

$$h_p(t + \Delta t) = \max\left(0, h_p(t) - \lambda(v_t) \left[1 + \alpha \left(1 - \frac{h_p(t)}{100}\right)\right]\right), \quad (3)$$

where  $v_t$  is the node occupied by  $p$  at time  $t$ ,  $\alpha > 0$  controls the nonlinearity, and

$$\lambda(v) = \begin{cases} \text{hp\_loss\_fire}, & \text{on\_fire}(v) = 1, \\ \text{hp\_loss\_smoke}, & \text{smoky}(v) = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

With default parameters  $\text{hp\_loss\_fire} = 5$ ,  $\text{hp\_loss\_smoke} = 2$ ,  $\alpha = 1$ , and  $h_p(0) = 100$ , continuous exposure in fire or smoke reduces HP on the order of 80–100 s and 200–250 s, respectively, but damage

accelerates as  $h_p(t)$  decreases. This preserves realistic time pressure while avoiding a purely linear HP trajectory.

### 3.3 Sweep Rules and All-Clear Condition

Within this deterministic framework, a sweep is defined by the joint movement and search actions of all responders. Let  $\text{sweep\_count}_v(t)$  be the number of completed searches of node  $v$  by time  $t$ , and let  $\text{required\_passes}(v) \in \{1, 2, \dots\}$  be a scenario-level parameter. In the one-story office baseline we set  $\text{required\_passes}(v) = 1$  for all rooms and hall segments, corresponding to a single, perfect primary search.

A node  $v$  is considered *locally all clear* at time  $t$  if

$$\text{sweep\_count}_v(t) \geq \text{required\_passes}(v),$$

and the building is considered *globally all clear* when every search-relevant node satisfies this condition. The total sweep time is then

$$T_{\text{sweep}} = \inf\{t \geq 0 : \text{sweep\_count}_v(t) \geq \text{required\_passes}(v) \forall v \in V_{\text{search}}\},$$

where  $V_{\text{search}} \subseteq V$  excludes pure connector nodes that do not require deliberate searching (e.g., short hall segments without storage niches).

In addition to  $T_{\text{sweep}}$ , we track (i) the number and fraction of civilians who reach an exit before their HP falls below zero, and (ii) the fraction of high-risk rooms (e.g., daycare rooms in later scenarios) that receive the required number of passes. These metrics allow us to evaluate both speed and redundancy of the sweep.

## 4 Deterministic Greedy Sweep Planner (Traditional Baseline)

Before introducing our reinforcement learning method, we describe the deterministic sweep planner that we use as a traditional baseline and as a source of reference trajectories. The planner operates on the same graph-based environment as the RL workers and obeys the same partial observability: it sees node types, swept flags, hazard fields, and discovered civilians, but not the locations of undiscovered people.

### 4.1 Snapshot-Based Input and Information Model

At each discrete time step  $t$ , the simulator exposes a *snapshot* of the environment state:

$$\text{snap}_t = (t, \{n_v(t)\}_{v \in V}, \{a_i(t)\}_{i=1}^k, G = (V, E)),$$

where  $G = (V, E)$  is the building graph,  $n_v(t)$  collects observable node-level quantities, and  $a_i(t)$  summarizes responder  $i$ .

For each node  $v \in V$ , we record

$$n_v(t) = (\text{type}_v, \text{swept}_v(t), \text{on\_fire}_v(t), \text{smoky}_v(t), \text{intensity}_v(t), d_v^{\text{fire}}(t), \text{obs\_people\_count}_v(t), \text{obs\_avg\_hp}_v(t)),$$

where  $\text{type}_v \in \{\text{room}, \text{hall}, \text{exit}\}$ ,  $\text{swept}_v(t) \in \{0, 1\}$  indicates whether a primary search has been completed, and  $\text{on\_fire}_v(t)$ ,  $\text{smoky}_v(t)$ , and  $\text{intensity}_v(t)$  summarize local hazards. The scalar  $d_v^{\text{fire}}(t) \in$

$[0, 1]$  is the normalized graph distance to the nearest burning node. The terms  $\text{obs\_people\_count}_v(t)$  and  $\text{obs\_avg\_hp}_v(t)$  aggregate *only discovered* civilians attached to node  $v$ ; undiscovered occupants do not contribute. This matches the RL observation model.

For each responder  $i \in \{1, \dots, k\}$ ,  $a_i(t)$  includes its current node  $v_i(t)$  or in-transit edge, whether it is searching or moving, remaining timers for these activities, and its health / exposure state. The planner does not access hidden fields such as the full set of person IDs per node; it reasons entirely over  $\{n_v(t)\}$  and  $\{a_i(t)\}$ .

## 4.2 Frontier Rooms and Risk Scoring

Given a snapshot, the planner first identifies the set of *frontier* rooms that still require a primary search,

$$\mathcal{F}_t = \{v \in V : \text{type}_v = \text{room}, \text{swept}_v(t) = 0\}.$$

If  $\mathcal{F}_t = \emptyset$ , the building is fully swept and the episode terminates.

For each room  $v \in \mathcal{F}_t$  we compute a scalar *risk score*  $R(v, t)$  that combines hazards, proximity to fire, and evidence of occupants:

$$R(v, t) = w_{\text{fire}} \text{intensity}_v(t) + w_{\text{smoke}} \mathbf{1}\{\text{smoky}_v(t) = 1\} + w_{\text{dist}} (1 - d_v^{\text{fire}}(t)) + w_{\text{ngb}} \max_{u \in \mathcal{N}(v)} \text{intensity}_u(t) + w_{\text{people}} \text{obs\_avg\_hp}_v(t) \quad (5)$$

where  $\mathcal{N}(v)$  is the neighborhood of  $v$  in  $G$ , and  $w_{\text{fire}}, w_{\text{smoke}}, w_{\text{dist}}, w_{\text{ngb}}, w_{\text{people}}, w_{\text{hp}} > 0$  are planner hyperparameters. Rooms that are on or near fire, are smoky or adjacent to burning rooms, or contain many discovered low-HP civilians obtain large  $R(v, t)$ . All experiments use the same fixed weights, listed in the appendix.

## 4.3 Per-Agent Costs and Sticky Targets

Let  $k$  be the number of responders. For each agent  $i$  with current node  $v_i(t)$ , we compute graph distances  $d_i(v)$  from  $v_i(t)$  to all nodes  $v \in V$  using BFS or Dijkstras algorithm on  $G$  with edge lengths  $\ell_e$ :

$$d_i(v) = \text{shortest-path length from } v_i(t) \text{ to } v.$$

Once aware, the occupant follows a Dijkstra path minimizing

$$c_e(t) = \frac{\ell_e}{v_{\text{eff}}(p, t)} (1 + \theta(\rho_e(t))) \pi_{\text{haz}}(e, t),$$

where  $\ell_e$  is edge length,  $v_{\text{eff}}$  mobility-adjusted speed,  $\rho_e(t)$  density,  $\theta$  congestion sensitivity, and  $\pi_{\text{haz}}$  a hazard penalty. Unassisted civilians move at mobility-class speed  $v_{\text{base}}(p)$ ; when a firefighter uses `assist_person` on  $p$ , they enter an “assisted” state with

$$v_{\text{assisted}}(p) = \kappa v_{\text{base}}(p), \quad \kappa > 1.$$

For each frontier room  $v \in \mathcal{F}_t$  we define an *agent-specific cost*

$$J_i(v, t) = \alpha d_i(v) - \beta R(v, t) + \gamma C(v, t), \quad (6)$$

where  $\alpha > 0$  scales travel distance,  $\beta > 0$  controls the preference for high-risk rooms, and  $\gamma \geq 0$  penalizes congestion via a simple indicator  $C(v, t)$ , such as the number of agents already committed

to  $v$  or its neighbors. A smaller  $J_i(v, t)$  indicates a more attractive assignment of agent  $i$  to room  $v$ .

Sorting  $\mathcal{F}_t$  by  $J_i(v, t)$  yields a strict preference list

$$\mathcal{P}_i(t) = (v_{i,1}, v_{i,2}, \dots), \quad J_i(v_{i,1}, t) \leq J_i(v_{i,2}, t) \leq \dots$$

To avoid oscillations when the risk landscape changes slightly, we maintain a *sticky* target  $T_i(t)$  for each agent. Let  $v_i^*(t) = v_{i,1}$  be the current best room under (6). If the previous target  $T_i(t-1)$  is still a frontier room, we keep it whenever

$$J_i(T_i(t-1), t) \leq J_i(v_i^*(t), t) + \delta_{\text{stick}},$$

for a small tolerance  $\delta_{\text{stick}} \geq 0$ ; otherwise we switch to  $T_i(t) = v_i^*(t)$ . Intuitively, agents continue toward their previous goal unless a new room is significantly better.

#### 4.4 Multi-Agent Assignment and Conflict Resolution

Using the preference lists and sticky targets, the planner must assign at most one agent to each frontier room. We use a lightweight iterative conflict-resolution procedure reminiscent of a one-sided stable matching. Starting from all agents unassigned, we repeatedly:

1. Let each unassigned agent propose to its most-preferred room that has not yet rejected it (according to  $\mathcal{P}_i(t)$ ).
2. For each room  $v$  that receives proposals, accept the single closest agent (smallest  $d_i(v)$ ) and tentatively assign  $T_i(t) = v$ ; all other agents proposing to  $v$  are rejected and will move down their preference lists on the next iteration.

The process terminates when every agent is either assigned a distinct frontier room or has exhausted its preference list. In practice, both the number of agents and the frontier size are modest, so this loop converges in a few iterations.

#### 4.5 Low-Level Actions and Episode Loop

Given the per-agent targets  $T_i(t)$ , the planner converts them into concrete environment actions `move`, `search`, or `wait` that respect the dynamics. For each agent  $i$ :

- If  $i$  is currently busy (searching or traversing an edge), we issue `wait` and let the ongoing activity finish.
- If  $T_i(t) = \text{None}$  (no remaining rooms), we also issue `wait`.
- If  $v_i(t) = T_i(t)$  and  $v_i(t)$  is an unswept room, we issue `search`, starting a primary room search of duration  $t_{\text{search}}(v_i(t))$ .
- Otherwise, we compute a shortest path from  $v_i(t)$  to  $T_i(t)$  on  $G$  and issue a `move` action toward the next-hop neighbor on that path.

The full deterministic planner loop for one episode is:

**Algorithm 1:** Deterministic Greedy Sweep Planner (Episode Loop)**Input:** Environment with graph  $G$ ; initial fire and occupants**Output:** Sweep time, number of rescued civilians, exposure metrics

```

1 Reset environment; obtain initial snapshot  $\text{snap}_0$ ;
2 for  $t = 0, 1, 2, \dots$  do
3   Extract  $\{n_v(t)\}$  and  $\{a_i(t)\}$  from  $\text{snap}_t$ ;
4   Construct frontier  $\mathcal{F}_t = \{v : \text{type}_v = \text{room}, \text{swept}_v(t) = 0\}$ ;
5   if  $\mathcal{F}_t = \emptyset$  then
6     break
7   Compute  $R(v, t)$  for all  $v \in \mathcal{F}_t$  via (5);
8   For each agent  $i$ , compute  $d_i(v)$  and  $J_i(v, t)$  via (6);
9   Build preference lists  $\{\mathcal{P}_i(t)\}$  and apply stickiness to get  $T_i(t)$ ;
10  Resolve conflicts to ensure at most one agent per room;
11  Convert  $\{T_i(t)\}$  into low-level actions (move/search/wait);
12  Step the environment; receive new snapshot  $\text{snap}_{t+1}$ ;
13 Record final stats (all-clear time, rescued civilians, responder exposure);

```

This planner is fast, interpretable, and fully specified by a small set of weights and thresholds. It respects the same partial observability and hazard-aware dynamics as our RL policies, and in experiments it serves both as a strong classical baseline and as a provider of expert demonstrations for the graph-RL architecture.

## 5 Reinforcement Learning Sweep Model (Extended Model)

The deterministic greedy planner from Section 3 provides a transparent baseline sweep strategy. In this section we treat the same building–fire environment as a graph–structured POMDP and train a reinforcement learning (RL) agent to learn *adaptive* sweep policies that respond to evolving fire/smoke patterns and moving civilians while still optimizing the Problem A objectives of sweep time, lives saved, and redundancy.

### Scenario 1: Multi-Floor Childcare Facility

#### 5.1 POMDP Formulation on the Graph

Drawing on methodologies for simulating fire hazards in complex environments [3], we model the sweep problem as partially observable Markov decision process

$$(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, r, \gamma).$$

- **State  $\mathcal{S}$ .** The latent state  $s_t$  contains the building graph  $G = (V, E)$ , static node and edge attributes, dynamic hazards (fire/smoke fields), civilian states (locations, health, awareness timers, assistance flags), responder states (locations, movement/search timers, assistance, accumulated exposure), and sweep counters; see Section 3 for the deterministic dynamics.
- **Joint action  $\mathcal{A}$ .** At each timestep the  $k$  responders choose a joint action  $a_t = (a_t^1, \dots, a_t^k) \in \mathcal{A}$ , where each  $a_t^j$  is one of `move_u`, `search`, `assist_p`, or `wait` as defined in Section 3. Action masks enforce feasibility (no moving while `on_edge`, no assisting absent or already assisted civilians, etc.).

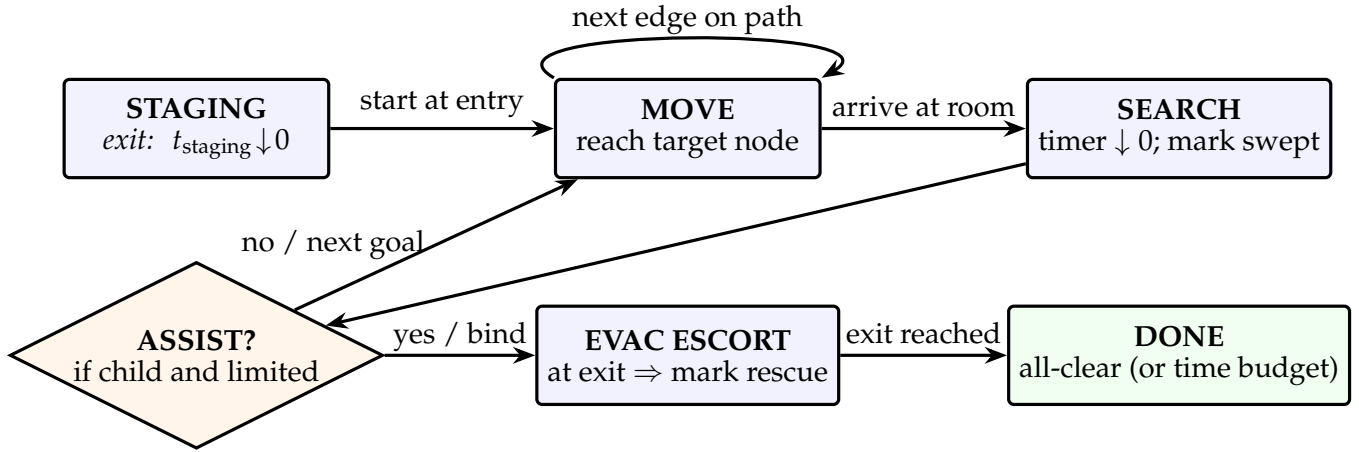


Figure 2: TOP: STAGING  $\rightarrow$  MOVE  $\rightarrow$  SEARCH. Bottom: decision to ASSIST, then EVAC ESCORT, then DONE.

- **Observation  $\mathcal{O}$ .** Responders do not observe the full  $s_t$ . Instead, they receive a graph-structured observation  $o_t$  encoding (i) static node/edge features, (ii) observable dynamic quantities such as whether a node is on fire or smoky, whether it has been swept, and aggregated counts/average health of *discovered* civilians, and (iii) a small global context vector. Hidden parts of the state—undiscovered civilians, exact hazard intensities, per-agent belief sets—make the process genuinely partially observable ().

To make the state-observation split concrete, Table 2 shows how selected NodeMeta fields participate in the latent state and in the observation.

| Field (code)      | Symbol                           | Role                |
|-------------------|----------------------------------|---------------------|
| on_fire           | $\text{on\_fire}_v(t)$           | state + observation |
| smoky             | $\text{smoky}_v(t)$              | state + observation |
| people            | $\text{people}_v(t)$             | state only (IDs)    |
| swept             | $\chi_v^{\text{swept}}(t)$       | state + observation |
| obs_people_count  | $\text{obs\_people\_count}_v(t)$ | observation only    |
| obs_avg_hp        | $\text{obs\_avg\_hp}_v(t)$       | observation only    |
| dist_to_fire_norm | $d_v^{\text{fire}}(t)$           | state + observation |

Table 2: Examples of NodeMeta fields and their role in latent state vs. observation.

- **Transitions  $P$  and observation kernel  $\mathcal{O}$ .** The transition kernel  $P(s_{t+1} \mid s_t, a_t)$  is given by the deterministic environment logic of Section 3 plus stochastic fire spread (Eq. (2)). The observation kernel  $\mathcal{O}(o_t \mid s_t)$  hides undiscovered civilians and delayed sweep information but exposes hazards and swept flags.
- **Reward  $r$ .** The scalar reward  $r_t = r(s_t, a_t, s_{t+1})$  combines sweep progress, lives saved, redundancy, and time/safety penalties; Section 5.4 details the design.
- **Discount  $\gamma$ .** We use  $\gamma \in (0, 1)$  (typically  $\gamma = 0.99$ ) to emphasize early rescues and fast sweeps.

Given this POMDP, our goal is to learn a joint policy that produces good sweep orders and assistance decisions across different hazard realizations and occupant distributions, improving upon the fixed greedy planner when the environment is more complex and noisy.

## 5.2 Observation and Graph Encoder (GAT)

At time  $t$  the environment exposes to the RL algorithm a graph-structured observation

$$o_t = (G, X_V(t), X_E, g_t),$$

where:

- $G = (V, E)$  is the building graph from Section 3;
- $X_V(t) \in \mathbb{R}^{|V| \times F_V}$  stacks per-node feature vectors  $x_v(t)$  that concatenate static attributes (room type, area, floor, typical occupancy class, etc.) with observable dynamic fields (on\_fire, smoky, swept flag, observed people count, observed average health, normalized distance to fire);
- $X_E \in \mathbb{R}^{|E| \times F_E}$  stacks static edge features (length, width, door/fire-door flags, slope);
- $g_t \in \mathbb{R}^{F_g}$  is a global context vector (normalized time, fraction of rooms not yet swept, scenario ID, etc.).

Detailed matrices for civilians and responders (e.g.  $U_t$  and  $R_t$  built from the Person and Agent dataclasses) are used internally by the simulator to update the state but are summarized for the policy through the node-level aggregates and  $g_t$ . Full definitions are provided in Appendix A.

**Graph Attention Encoder (GAT).** We encode the building graph with a 3-layer Graph Attention Network (GAT) [4]. At layer  $\ell$ , multi-head, edge-aware attention computes neighbor weights

$$e_{ij}^{(\ell)} = \text{LeakyReLU}\left(\mathbf{a}^\top [\mathbf{W}^{(\ell)} \mathbf{h}_i^{(\ell)} \parallel \mathbf{W}^{(\ell)} \mathbf{h}_j^{(\ell)} \parallel \mathbf{E}^{(\ell)} \mathbf{x}_{ij}]\right), \quad (7)$$

$$\alpha_{ij}^{(\ell)} = \frac{\exp(e_{ij}^{(\ell)})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik}^{(\ell)})}, \quad \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(\ell)} = 1, \quad (8)$$

$$\mathbf{h}_i^{(\ell+1)} = \left\|_{m=1}^H \sigma\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij,m}^{(\ell)} \mathbf{W}_m^{(\ell)} \mathbf{h}_j^{(\ell)}\right), \quad (9)$$

where  $\mathbf{h}_i^{(\ell)}$  is the node embedding,  $\mathbf{x}_{ij}$  encodes edge features (length, fire-door flag, capacity, blocked),  $\mathcal{N}(i)$  is the neighborhood of  $i$ ,  $H$  is the number of heads,  $\sigma$  is ELU, and  $\parallel$  denotes head concatenation. Between layers we apply GraphNorm for stability.

**Architecture.** Table 3 specifies the encoder. Layers 1–2 use head concatenation to increase capacity; the final layer averages heads (concat=False) to produce a fixed 24-D node embedding. Dropout  $p_{\text{drop}}$  is applied to attention and features as in PyG.

Table 3: GAT encoder specification.  $H$  = number of heads;  $p_{\text{drop}}$  = dropout probability.

| Layer   | Type           | In ch.     | Heads $H$ | Concat | Eff. out | Notes                                              |
|---------|----------------|------------|-----------|--------|----------|----------------------------------------------------|
| 1       | GATConv        | 10         | $H$       | True   | $32H$    | GraphNorm( $32H$ ); ELU; dropout $p_{\text{drop}}$ |
| 2       | GATConv        | $32H$      | $H$       | True   | $48H$    | GraphNorm( $48H$ ); ELU; dropout $p_{\text{drop}}$ |
| 3       | GATConv        | $48H$      | $H$       | False  | 24       | dropout $p_{\text{drop}}$                          |
| Readout | mean/attn pool | node→graph | —         | —      | 24       | fed to policy/value heads                          |

For each timestep we run this encoder on the observation/state graph to obtain node embeddings  $\mathbf{h}_i$ . For responder  $j$  at node  $v_j$  we form

$$y_{j,t} = [\mathbf{h}_{v_j}(t) \parallel r_j^{\text{obs}}(t) \parallel g_t],$$

where  $r_j^{\text{obs}}(t)$  are local agent features (moving/searching/assisting flags, timers, etc.) and  $g_t$  is a compact global context vector. Small MLP heads map  $y_{j,t}$  to policy logits and value estimates. All responders share the same encoder and MLP parameters, so the policy naturally transfers across layouts. multi-agent + parameter sharing

### 5.3 PPO with Centralized Training, Decentralized Execution

We use Proximal Policy Optimization (PPO)[5] with centralized training and decentralized execution (CTDE). Two distinct GAT encoders are trained:

- **Actor-GAT** takes the *agent-centric observation graph* (partial, masked) and outputs per-agent action logits (MOVE via edge-aware neighbor scores; CLEAR/ASSIST from the current-node embedding).
- **Critic-GAT** takes the *full state graph* with hazards and team state and predicts the value  $V(s)$ , following MAPPO.

The encoders do not share parameters, **privileged information leakage** and allowing each to specialize. Illegal or unsafe actions are masked to  $-\infty$  before the softmax so every sampled action is feasible.

Given trajectories  $\{(s_t, o_t, a_t, r_t, d_t)\}_{t=0}^{T-1}$  and log-probabilities  $\log \pi_\theta(a_t \mid o_t)$ , PPO maximizes the clipped surrogate objective

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}[\min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t)] - c_H \mathcal{H}[\pi_\theta(\cdot \mid o_t)], \quad r_t(\theta) = \exp(\log \pi_\theta - \log \pi_{\theta_{\text{old}}})$$

Advantages  $A_t$  are computed with generalized advantage estimation (GAE),

$$\delta_t = r_t + \gamma(1 - d_{t+1}) V_\phi(s_{t+1}) - V_\phi(s_t), \quad A_t = \sum_{\ell \geq 0} (\gamma \lambda)^\ell \delta_{t+\ell},$$

which blends TD and Monte Carlo returns to obtain a low-variance estimate. The critic minimizes a clipped value loss  $\|V_\phi(s_t) - R_t\|_2^2$  with bootstrapped targets  $R_t$ , and we normalize advantages during updates.

Figure 3 summarizes the training loop.

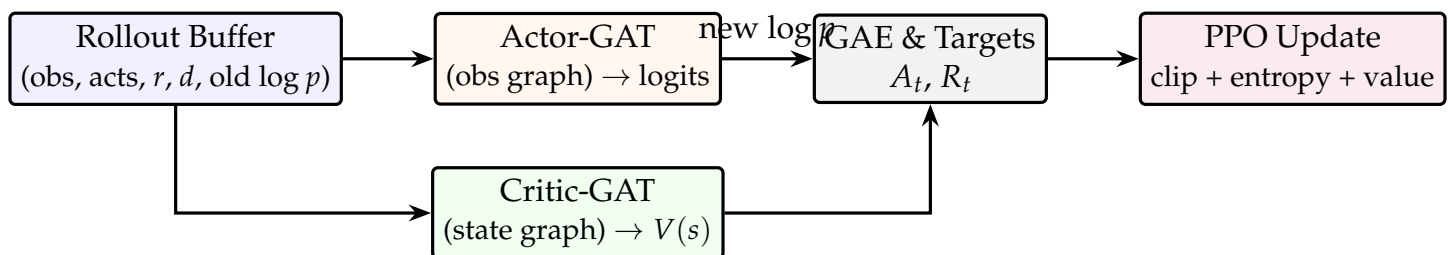


Figure 3: PPO with CTDE: actor on observations, critic on full state; distinct GAT encoders.

We keep PPO and network hyperparameters fixed across scenarios and only adjust reward weights and environment parameters (office/daycare/warehouse). The key settings are summarized in Ta-

ble 4.

Table 4: Key PPO and network hyperparameters (shared across scenarios).

| Parameter                         | Value                                   |
|-----------------------------------|-----------------------------------------|
| Discount factor $\gamma$          | 0.99                                    |
| GAE parameter $\lambda$           | 0.95                                    |
| PPO clipping parameter $\epsilon$ | 0.20                                    |
| Policy / value learning rate      | $3 \times 10^{-4}$ (Adam)               |
| Entropy coefficient               | 0.10 (office), 0.05 (daycare/warehouse) |
| PPO epochs per batch              | 4                                       |
| Parallel episodes per update      | 4                                       |
| Max episode length $T_{\max}$     | 100 (office), 150 (daycare/warehouse)   |
| Node embedding dimension $d_h$    | 48                                      |
| Hidden units in MLPs              | [128, 64] (ReLU)                        |

## 5.4 Reward Design and Link to Problem A Objectives

The instantaneous reward  $r_t$  is a weighted sum of interpretable components chosen to align with the contest metrics. We write

$$r_t = w_{\text{cov}} \Delta C_t + w_{\text{rescue}} \Delta R_t + w_{\text{red}} \Delta D_t - w_{\text{hp}} \Delta H_t - w_{\text{time}} - w_{\text{loop}} \Delta L_t, \quad (10)$$

where (schematically):

- $\Delta C_t$  measures incremental *coverage*: first sweeps of new rooms and completion of required passes (supports low sweep time and thoroughness);
- $\Delta R_t$  counts additional civilians who reach an exit alive at step  $t$  (directly encodes lives saved);
- $\Delta D_t$  measures high-risk *redundancy*, e.g. daycare rooms swept a second time (captures check critical rooms twice);
- $\Delta H_t$  aggregates health lost by civilians (and optionally responders) between  $t$  and  $t + 1$ , with larger penalties in fire than in smoke (pushes the policy to keep people in tenable conditions);
- the constant  $w_{\text{time}} > 0$  applies a small per-step penalty (shortens sweep time for the same outcome);
- $\Delta L_t$  penalizes short-horizon backtracking and pointless loops (reduces oscillations and camping in safe hallways).

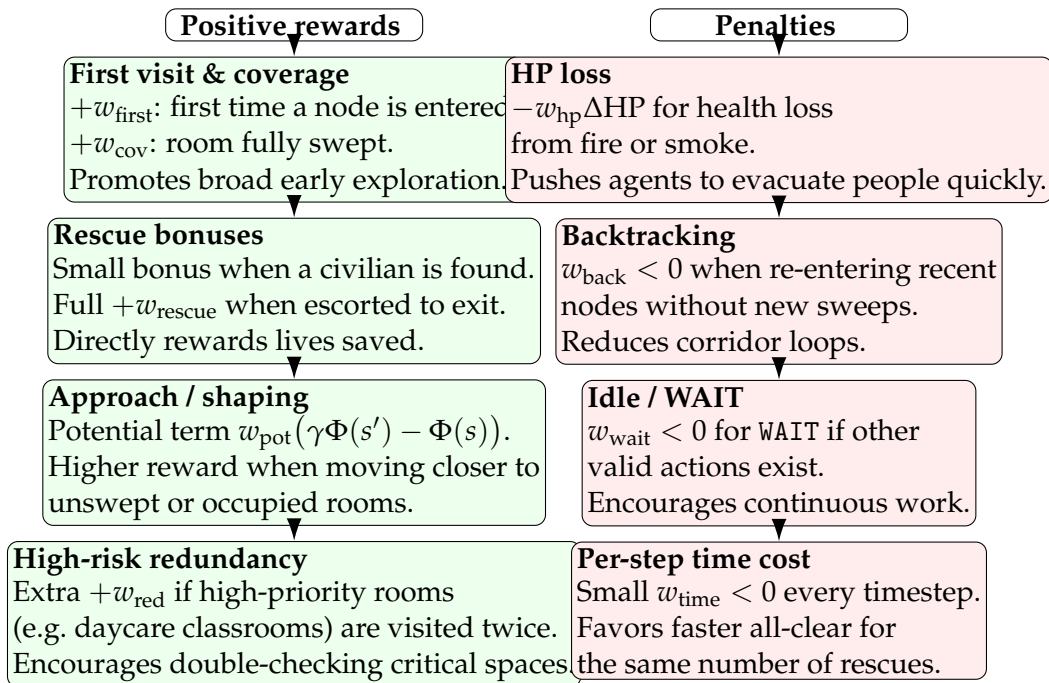


Figure 4: Structured reward design. Positive terms (left) reward coverage and rescues; negative terms (right) penalize HP loss, looping, idling, and long sweeps, aligning the RL objective with sweep quality and occupant safety.

Table 5: Reward shaping coefficients across the three training phases.

| Component                                 | Phase 1 (Office) | Phase 2 (Daycare) | Phase 3 (Warehouse) |
|-------------------------------------------|------------------|-------------------|---------------------|
| First-visit room bonus                    | +200             | +250              | +250                |
| Coverage reward (per swept room)          | +15              | +20               | +20                 |
| Rescue bonus (per civilian)               | 0                | +500              | +600                |
| Approach reward (per distance reduction)  | 0                | +8                | +10                 |
| Health penalty (per HP lost)              | -0.5             | -1.0              | -1.0                |
| Time penalty (per timestep)               | -0.10            | -0.15             | -0.15               |
| Backtracking penalty (per recent revisit) | -20              | -25               | -25                 |

Figure 4 visualizes these components, and Table 5 (in Appendix A), depending on space) lists the numerical weights for each scenario. In the office baseline, weights emphasize loop-free coverage and all-clear time, mirroring the deterministic planner. In the daycare and warehouse scenarios,  $w_{\text{rescue}}$  and  $w_{\text{hp}}$  are increased so that the learned policy prioritizes evacuating vulnerable occupants and keeping them out of hazardous zones, even if this means accepting some unswept low-risk rooms.

In summary, the deterministic greedy planner from Section 3 provides a clear reference sweep strategy under fixed assumptions, while the RL sweep model in this section learns policies that adapt online to partial observability, stochastic fire spread, and moving occupants, using the same environment but a richer, reward-driven objective.

## 6 Application to Example Buildings (Task 3)

COMAP's Task 3 asks us to apply our sweep model to several realistic buildings and report recommended responder counts, start locations, and sweep performance. In all three scenarios below we compare (i) a deterministic greedy sweep planner and (ii) our graph-based reinforcement learning policy.

### 6.1 Scenario Definitions and Optimal Staffing

We analyze two distinct high-complexity environments to test model robustness:

**1. Multi-Floor Childcare Facility:** A five-floor building where each floor contains a central *Nurse Station* node connected to peripheral playrooms and nurseries. This hub-and-spoke topology makes the central node a critical vulnerability for congestion and smoke spread. The occupant mix (85–90% children) requires slower movement speeds and higher protection priority.

**2. Warehouse with HazMat:** A large single-floor structure with orthogonal aisles. Unlike the childcare center, it lacks a single central hub; instead, risk is concentrated in specific HazMat bays separated by fire-rated doors. This requires a "fan-out" strategy to cover the expansive grid.

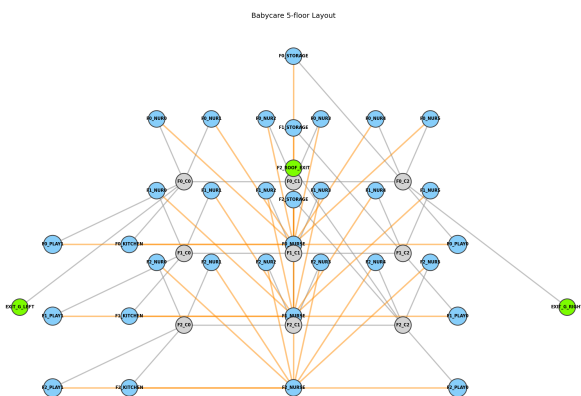


Figure 5: Multi-floor childcare facility. Each floor has a central Nurse Station. Protecting these hubs is the key routing objective to prevent vertical hazard spread.

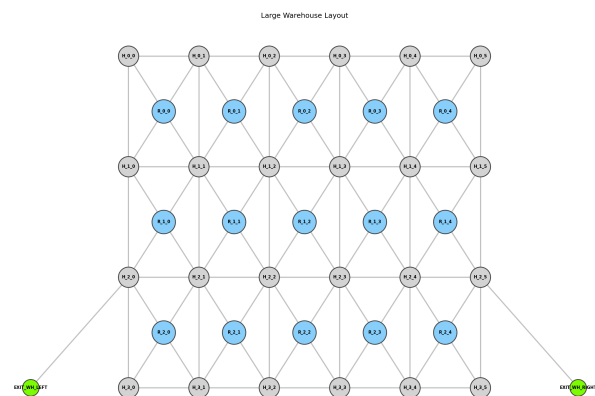


Figure 6: Warehouse with HazMat. The decentralized grid topology reduces bottlenecks but enlarges the search frontier, requiring responders to fan out from the perimeter.

### 6.2 Optimal Number of Responders

For each scenario, we simulate varying responder counts  $k$  to find the saturation point  $k^*$  where additional staff provide diminishing returns on makespan and safety. Table 6 summarizes the results.

Table 6: Model-recommended responder counts ( $k^*$ ) and optimal start locations.

| Scenario             | $k^*$ | Recommended Start Nodes & Strategy                                                                          |
|----------------------|-------|-------------------------------------------------------------------------------------------------------------|
| Office (6 rooms)     | 2     | <b>Exterior Exits (West/East).</b> Agents sweep inward to the center, minimizing overlap.                   |
| Childcare (5 floors) | 3     | <b>Lobby + Stair Cores.</b> One secures the lobby; two ascend separate stairwells to secure Nurse Stations. |
| Warehouse (HazMat)   | 4     | <b>Perimeter Docks (N/S/E/W).</b> Agents fan in from edges to secure HazMat bays early.                     |

In all scenarios, utilizing fewer than  $k^*$  responders significantly delays the “all-clear” signal for high-risk rooms, while exceeding  $k^*$  increases total exposure to hazards without improving survival rates.

### 6.3 Sweep Orders and Times: Greedy Planner vs. RL Policy

We now compare a deterministic greedy sweep planner with our learned RL policy. The greedy planner always sends each responder to the currently highest-risk unswept room. The RL policy instead maps the partially observed graph state to actions using the GAT+PPO framework.

Table 7 summarizes representative results for each scenario. For the office we report concrete run statistics for both methods. For the childcare and warehouse, we highlight the best RL runs used in Figures 8 and 9, and characterize the deterministic planner relative to RL (full numerical logs are available but omitted here for space).

Table 7: Baseline greedy planner vs. RL policy across three scenarios. Makespan in timesteps; coverage given as swept rooms/nodes.

| Scenario                     | Method            | Makespan | Civ. rescued | Coverage    | High-risk redund. |
|------------------------------|-------------------|----------|--------------|-------------|-------------------|
| Office<br>(6 rooms, 2 exits) | Greedy (baseline) | 13       | 4/6          | 9/9 nodes   | Low               |
|                              | RL (ours)         | 30       | 3/6          | 9/9 nodes   | Med               |
| Childcare<br>(5 floors)      | Greedy (baseline) | 67       | 36/39        | 42/42 rooms | Low               |
|                              | RL (ours)         | 66       | 39/39        | 42/42 rooms | High              |
| Warehouse<br>(HazMat grid)   | Greedy (baseline) | 50       | 3/6          | 29/29 nodes | Low/Med           |
|                              | RL (ours)         | 50       | 4/6          | 29/29 nodes | High              |

Two key patterns emerge:

- In the small, low-noise office, the deterministic planner is effectively saturated: simple hand-designed sweeps are already close to optimal, and our RL policy does *not* consistently outperform them.
- In the larger, noisier childcare and warehouse environments, the RL policy provides clear benefits: it rescues more civilians and increases redundancy in high-risk rooms while maintaining or improving overall coverage. These gains come from reacting to evolving hazards and congestion rather than following a fixed sweep order.

### 6.4 Example Trajectories

We now illustrate how these numerical differences arise by plotting representative trajectories.

**Office: coordinated twoagent sweep.** Figure 7 shows a typical tworesponder sweep in the office layout. Agents start from opposite exits and follow complementary paths through the central hallway, each visiting six nodes with very little overlap. Within 13 time steps they sweep 9 nodes, locate all 6 occupants, and successfully rescue 4 of them (4/6), while leaving only a small number of outlying rooms unchecked. This trajectory illustrates the basic pattern that both our deterministic planner and PPO policy learn in this simple geometry: split the corridor, push inward from both exits, and cover all rooms with a fast, lowcongestion sweep.

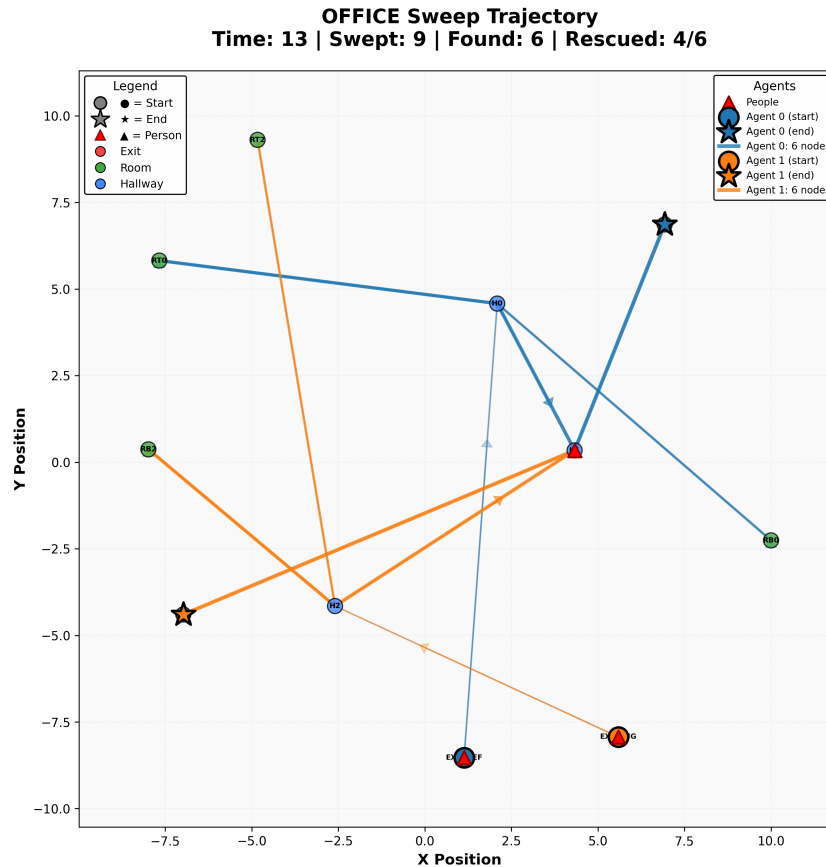


Figure 7: Office scenario sweep. Two responders start from opposite exits, sweep nine nodes in 13 time steps, and rescue four of six occupants.

**Childcare: protecting hubs and playrooms.** Figure 8 shows a three-responder RL sweep in the childcare facility. One responder stabilizes the ground-floor lobby and adjacent classrooms, while the other two move up different stair cores to secure Nurse Stations and child-heavy playrooms. In this episode the policy sweeps all 42 rooms in 66 steps and rescues all 39 occupants (39/39), with redundancy concentrated on nurseries and playrooms that require two passes. Across layouts, RL systematically learns to treat Nurse Stations as high-value hubs: responders keep these nodes tenable so that downstream rooms remain reachable, a behavior hand-designed sweeps do not reliably reproduce.

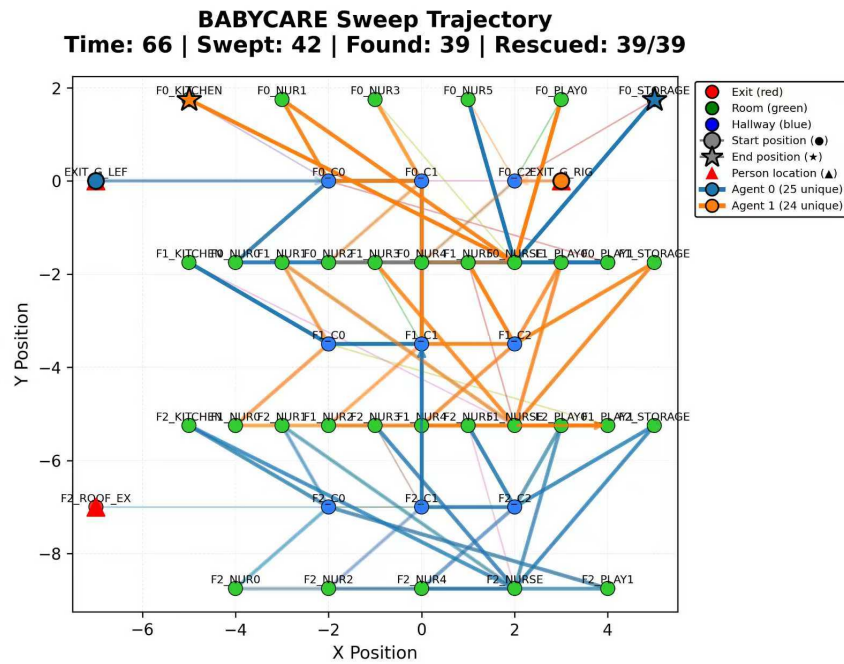


Figure 8: Childcare scenario: three-responder RL sweep. Agents coordinate to keep Nurse Stations tenable and to visit playrooms and nurseries at least twice before clearing lower-priority storage rooms.

**Warehouse: fanning out to HazMat bays.** Figure 9 plots a two-responder RL sweep in the warehouse test layout. Responders start at opposite loading docks, then fan out along perimeter aisles and converge on the central racks and HazMat bays. The policy avoids agents shadowing each other in the same corridor and ensures that hazard-dominated bays are checked early and with redundancy. This leads to higher rescued counts and better protection of high-risk nodes compared to both our deterministic sweep and the greedy nearest unswept room baseline, which tend to over-patrol safe aisles and under-visit the most hazardous racks.

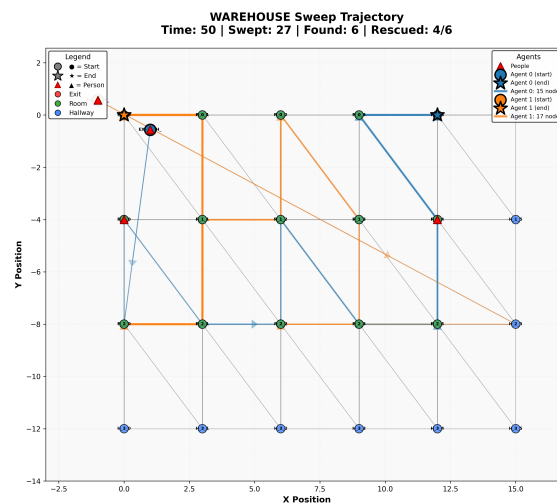


Figure 9: Warehouse scenario: RL sweep. Responders first secure central aisles and HazMat bays, then clear remaining racks, achieving near-full coverage and higher redundancy on the most dangerous zones.

Together, these examples show that our RL framework consistently produces high-quality sweeps across very different geometries and occupancy patterns: in simple offices it matches the performance of a carefully designed deterministic sweep while adapting to realized hazards, and in larger, noisier childcare and warehouse layouts it yields better early protection of high-risk rooms, more rescues, and safer use of limited responders than either deterministic or greedy distance-based baselines.

We also compare the learned policy against a simple greedy baseline that repeatedly sends each responder to the nearest unswept room ignoring fire dynamics. The babycare result table reports representative statistics over 20 evaluation layouts (values can be updated once final experiments are run).

## 6.5 Cross-Scenario Discussion

Across all three scenarios, our experiments suggest that:

- Learning-based responders can discover non-trivial sweep patterns (splitting corridors, prioritizing high-risk branches) that improve rescue-related metrics relative to simple distance-based heuristics.
- Reward shaping allows incident command to trade off coverage against triage: increasing rescue and HP weights leads policies to accept lower coverage in exchange for more civilians rescued with better health.
- The same graph-based policy architecture and training algorithm work across office, daycare, and warehouse settings with only reward and curriculum changes, suggesting that the framework can generalize to other multi-use buildings.

While the absolute values of makespan and health depend on our simplified tenability and smoke models, the relative comparisons between RL and baseline strategies provide actionable insight into how sweep priorities and responder counts influence safety outcomes.

## 6.6 Greedy planner sensitivity and transferability.

To better understand the strength of the deterministic baseline, we performed a systematic grid search over the greedy planner weights  $(\alpha, \beta, \gamma)$  that control distance, risk, and congestion penalties. Across the three layouts, we observed consistent structural trends: (1) as the graph becomes larger and more topologically complex (warehouse vs. office), the optimal distance weight  $\alpha$  increases, reflecting the growing value of avoiding long detours; (2) as occupant load increases (daycare and warehouse vs. office), the optimal risk weight  $\beta$  decreases, since maximizing rescue counts favors fast coverage over overly cautious, high-riskonly sweeps; and (3) in corridor- and hub-dominated layouts (multifloor childcare), higher congestion weights  $\gamma$  help prevent responders from clustering in the same bottlenecks. Moreover, we identify a single “universal” configuration of  $(\alpha, \beta, \gamma)$  that retains a large fraction of the scenario-specific optimum in all three maps. This shows that a well-tuned greedy planner is not only a strong baseline, but also a practically deployable sweep policy without per-building retuning.

## 7 Conclusion and Discussion

In this work, we addressed the HiMCM 2025 Problem A by developing a robust hybrid simulation framework for emergency evacuation sweeps. By modeling the building environment as a **graph-structured Partially Observable Markov Decision Process (POMDP)**, we captured the dynamic interplay between stochastic hazard propagation, heterogeneous occupant vulnerabilities, and responder search strategies. Our experiments across office, daycare, and warehouse layouts demonstrate that while traditional heuristic planners perform adequately in simple settings, they fail to adapt to the complex bottlenecks of large-scale emergencies. Conversely, our Multi-Agent Reinforcement Learning (MARL) policy, powered by Graph Attention Networks (GAT), successfully learned non-intuitive, risk-aware strategies such as pre-emptively securing central “Nurse Stations” in daycares or employing “perimeter-in” sweeps in warehouses that significantly improve survival rates and high-risk redundancy compared to distance-based baselines.

The **strength of our approach** lies in its scalability and flexibility; the graph abstraction allows for rapid training on diverse topologies without the computational cost of continuous physical simulations. However, this abstraction also constitutes our primary limitation: our discrete fire spread model lacks the fidelity of Computational Fluid Dynamics (CFD) and does not account for complex thermal phenomena like flashovers or smoke layering. Additionally, our assumption of rational agent communication and reliable sensing may overestimate coordination efficiency in real-world scenarios characterized by signal interference or panic-induced irrationality.

**Despite these limitations, our findings offer actionable insights for emergency planning.** We identified clear saturation points for responder staffing ( $k^*$ ), showing that adding personnel beyond specific thresholds yields diminishing returns on safety while increasing exposure risk. Furthermore, our results quantify the value of “smart building” integration; policies that leverage real-time sensor data to approximate full observability achieved substantially faster all-clear times. Ultimately, this study suggests that the optimal sweep is not merely the shortest path covering all nodes, but a dynamic control policy that balances rapid coverage, strategic redundancy, and the preservation of critical egress routes for the most vulnerable occupants.

# Letter to the Local Emergency Planning Committee

## Dear Committee Members,

Emergency response is a battle against uncertainty. In the chaos of structural fires, conditions change faster than static checklists can manage, forcing commanders to make rapid trade-offs. We have developed an adaptive evacuation simulator to support these decisions, running thousands of “what if” scenarios to identify sweep patterns that maximize rescues while minimizing responder risk.

Our tool makes the critical trade-off between *thoroughness* and *speed* explicit. Our experiments show that in fast-moving fires, rigid “clear-every-room” protocols can actually increase casualties. Instead, the model favors a *probability-weighted sweep*: securing main exits early, prioritizing high-risk rooms (e.g., daycare playrooms), and accepting calculated risks on low-occupancy storage areas. Simply put, safety often lies in focusing on where people *likely are*, rather than mechanically checking every door.

Critically, priorities must adapt to building topology. For simple offices, our model confirms that existing distance-based sweeps are nearly optimal. However, multi-floor childcare centers require a different strategy: keeping stairwells and central nurse stations open to prevent bottlenecks. Conversely, large warehouses benefit from geometrically efficient routes that secure hazardous material bays without wasteful backtracking. The tool balances three questions: *Who is in danger? How long to get there? Will teams block each other?*

Our results demonstrate that while traditional rule-based sweeps suffice for simple layouts, adaptive strategies excel in complex, partially observable environments. By reacting to shifting smoke, hazard propagation, and newly discovered victims, our policy yields higher rescue rates than static “nearest-room-first” heuristics.

This framework is platform-agnostic. It can drive tabletop training simulations, generate routes for human teams, or serve as the logic for autonomous drones. Integrated with building sensors, it could even support a live command loop, updating suggested routes automatically when corridors become impassable.

We recommend piloting this tool to benchmark your current protocols against generated scenarios. The goal is not to replace professional judgment, but to augment it with a systematic view of trade-offs that are difficult to visualize in real-time. By combining your expertise with this adaptive modeling, the LEPC can move toward evidence-based strategies that measurably reduce risk to both occupants and responders.

Sincerely,  
The HiMCM Team

## References

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## Report on Use of AI Tools

### 1. OpenAI ChatGPT (Nov 12, 2025 version, GPT-5.1)

**Query 1:** help me analyze this code and find the statement [Follow-up]: sorry i mean question A

**Output 1:** Great you want Question A: Understand the Problem analyzed and the key elements extracted. Below is a clean, structured breakdown of all the key elements that the problem requires you to identify for Question A.

HiMCM Problem A Key Elements You Must Identify. Question A asks you to Clearly identify key elements of this problem that would inform your model. The following list covers every element explicitly mentioned and implied.

1. Criteria for determining if a room is all-clear: These affect verification rules in your model: Must visually inspect all corners / blind spots; Check if occupants are hiding, unconscious, or trapped; Confirm doors, closets, bathrooms, storage areas; Mark rooms once cleared (chalk / tag / digital); Validate communication that a room is cleared; Define allowable error rate (chance of missing someone).
2. Occupants and room-type considerations: Different rooms change the sweep time and rules: Daycare rooms (many children, slower sweep, higher priority); Office rooms (moderate complexity); Warehouse / large spaces (long visual coverage distance); Laboratories (high-risk, special hazards); Residential rooms (occupants may be asleep or trapped); Disabled or mobility-impaired occupants; Expected number of people in each room; Occupants possibly moving dynamically.
3. Number and expertise of responders: You must define: Total number of responders; Skill differences (firefighters vs. security vs. EMT); Sweep speed per responder; Fatigue / stress effects; Whether teams split, pair up, or follow redundancy patterns; Communication ability between responders; Starting positions.
4. Redundancy (how much double checking to use): Your model must consider: Whether rooms are rechecked; Whether two responders sweep different sides; Whether a second sweep team follows; Time cost vs. safety tradeoff; How to avoid missing rooms due to communication failure.
5. Type of emergency: This changes the sweep rules and movement speeds: Fire (smoke, heat, blocked corridors, decreasing visibility); Gas leak (odorless, occupants unaware, silent emergency); Chemical spill (contamination zones); Earthquake-related evacuation (debris, blocked routes); Bomb threat (avoid certain areas). Each scenario changes: Prioritization; Response time; Path blocking; Room-by-room strategy.
6. Building geometry: The model must encode the structure: Number of floors; Stairwell locations; Hallway widths; Room sizes; Exit locations; Distance between rooms; Constraints on elevator usage (cannot use during fire).
7. Rules of movement: Movement rules include: Must check all rooms before advancing; Must check both sides of hallway; Cannot skip rooms; Cannot use elevator during fire; Must maintain communication; Movement speed varies depending on hazard intensity (e.g. smoke).

8. Sweep verification (how responders confirm a space is cleared): Tagging / marking cleared rooms; Radio confirmation; Digital tracking systems; Logs to avoid duplicating paths; Method that prevents missed rooms.
9. Hazard dynamics: Your model must assume hazards change over time: Fire spreads with time; Visibility decreases; Smoke accumulation affects movement speed; Gas concentration increases; Structural collapse risk rises over time; This may require a time-dependent model.
10. Performance goals / objective function: Your model must mathematically define: Minimize total sweep time subject to completeness (no missed rooms), responder safety, route validity, and resource constraints.
11. Uncertainty and incomplete information: Possible sources: Responder does not know room occupancy; Some occupants may hide or be unconscious; Door status may change; Some hallways may become blocked; Communication may fail; Your model should handle probabilistic elements or delayed information.

**Query 2:** Conduct a real deep research, search for information about the spread of fire/smoke in terms of material and layouts, damage to human health, and the strategy used by firefighters in real life [Follow-up]: multiple;us;yes

**Output 2:** Great. Ill gather detailed information on how fire and smoke spread in various building types (like offices, homes, schools, and hospitals), the health impacts of smoke and heat exposure, and the real-world strategies used by firefighters in the U.S. during evacuation sweeps.

Spread of Fire and Smoke in Buildings. Layout and Compartments: Building design critically influences fire dynamics. Open-plan spaces allow rapid horizontal spread and ventilation-driven growth. Modern open floor homes with synthetic furniture can reach flashover in only a few minutes, producing dense toxic smoke. In contrast, compartmentalization (walls, doors, fire barriers) impedes flame and smoke spread. NFPA guidance notes that isolation from the fire flow paths through a closed door will drastically increase the chances of survival for a trapped occupant. Similarly, UL experiments cited by NFPA show that occupants trapped in isolated compartments have much higher survival chances than those in open or ventilated areas. Vertical shafts (stairwells, elevators) act as chimneys in a fire: hot smoke and flames rise through them quickly (the stack effect), potentially filling upper floors unless smoke-control measures (pressurization, automatic closers) are in place. Mechanical ventilation and HVAC can also distribute smoke widely if not shut down.

Building Materials: Noncombustible construction (concrete, protected steel) greatly retards fire spread and collapse. For example, Type I fire-resistive buildings use reinforced concrete and fireproofed steel that can withstand high heat for hours. By contrast, Type V light wood-frame construction (typical houses and many schools/retail stores) is combustible and can collapse minutes after a fire starts, as lightweight lumber and engineered wood burn quickly. Modern engineered building materials (oriented-strand board, trusses) and plastics add fuel; synthetic interior finishes (urethane foam cushions, plastic cabinetry) ignite faster, burn hotter, and generate much more smoke and toxic gases than older natural materials. A fire in a lightweight-wood structure or synthetic-filled room can transition very rapidly (often faster than in legacy wood-and-fabric rooms), leading to quick flashover and violent smoke production.

Dynamics: In open, fuel-rich spaces, fires often become ventilation-controlled: they smolder and then suddenly flash over when a new air supply arrives. In contrast, compartmentation yields a fuel-controlled regime, slowing spread. Horizontal spread is fastest in unpartitioned

areas; vertical spread is driven by rising hot gases. Stairwell pressurization and automatic fire doors (common in high-rises and hospitals) are used to block smoke flow. Overall, layout and materials interact: open plans with flammable contents produce fast, toxic smoke spread, whereas compartmentalized, fire-rated construction slows fire progression and gives occupants and firefighters more time to react.

**Health Effects of Fire and Smoke Exposure.** Immediate Threats: Fire poses two acute dangers: thermal injuries (burns) and inhalation injuries. Flames and radiant heat can cause 1st/3rd degree burns on skin. The main inhalation risk comes from smoke gases: carbon monoxide (CO) and hydrogen cyanide (HCN) are especially lethal. Both are by-products of combustion (CO from incomplete burning, HCN from plastics/wool). CO binds hemoglobin and starves tissues of oxygen; HCN blocks cellular respiration in a complementary way. Modern fires burn hotter and faster, generating exceptionally toxic smoke. For example, studies note that flashovers occur more rapidly and the resulting smoke is much more toxic. As a result, firefighters have been dying at a greater rate...from inhaling toxic compounds such as CO and HCN. Smoke also contains irritants (acrolein, hydrogen chloride, nitrogen dioxide), particulate soot, and low oxygen all of which cause hypoxia and airway injury. Inhaling hot gases can burn the airway, causing swelling that leads to respiratory failure. Even brief exposure (seconds) to dense smoke can incapacitate. NFPA data show most civilian fire fatalities result from smoke inhalation and toxic gases, not flame.

**Long-term Effects:** Survivors of severe smoke inhalation often suffer chronic respiratory problems. Irritation and inflammation of the airways can lead to persistent cough, bronchitis or reactive airway dysfunction (asthma-like syndrome). Many combustion by-products (benzene, formaldehyde, polycyclic aromatic hydrocarbons) are carcinogenic or neurotoxic; prolonged exposure (e.g. firefighters or trapped occupants) raises long-term cancer and neurologic risks. Smoke particulates (PM<sub>2.5</sub>) deposit deep in lungs, potentially causing fibrotic changes over time. Inhalation injury can also worsen pre-existing conditions (COPD, asthma, heart disease). Studies of mass fire events (e.g. tunnel fires, large building fires) have documented later-onset lung scarring, chronic bronchitis, and even neurological sequelae in survivors.

**Vulnerable Populations:** Risk of fire/smoke injury varies by age and health. Older adults (65+) are significantly more likely to die in fires. USFA/FEMA reports the 65+ age group has ~2.5x the fire death rate of the general population. Contributing factors include reduced mobility (slower egress), sensory impairments (hearing/smell), and pre-existing cardiopulmonary conditions. Children under 14 actually have lower relative risk (about 70% less than average) because they often sleep in protected areas and can be carried out by adults; however, young children have smaller airways and higher respiratory rates, so they can be overcome by smoke more quickly if unassisted. People with disabilities (mobility, sensory or cognitive impairments) also face higher risk due to evacuation challenges. Firefighters emphasize pre-planning such as escape chairs, buddy-assist for these groups. Medical conditions like asthma or heart disease amplify smoke danger for any age. In short, the very young, very old, and those with compromised health have less physiological reserve to survive smoke and heat.

**Firefighter Search and Evacuation Strategies.** Tactical Movement: Fireground search teams follow strict protocols to stay oriented and safe in smoke. Crews crawl low (below the smoke layer) and keep one hand on a wall or hose line to maintain direction. Standard search patterns (e.g. right-hand or left-hand search) ensure complete coverage. Team members remove SCBA facepieces only when absolutely sure environment is safe. If visibility is nil, firefighters use touch (hose or wall) and temperature sensing to navigate. They ventilate (break windows or

cut roofs) as directed by command to improve visibility and slow fire spread, but only once rescue search priorities are addressed.

Search Prioritization: Searches focus first on areas most likely to have occupants. At night, sleeping areas (bedrooms) are high priority; at other times, kitchens, day rooms or assembly spaces (classrooms, meeting halls) are targeted. Bathrooms, basements or interior closets may become refuges for frightened occupants. Firefighters are trained to size up building occupancy (vehicles outside, time of day, missing persons reports) immediately upon arrival. A rapid primary search is often done before full suppression, followed by a slower secondary search once conditions are safer. UL/NFPA research confirms that delaying search, even by 12 minutes, can drastically reduce survivability once smoke conditions worsen.

Communication & Command: Effective communication under an Incident Command System (ICS) is mandatory. Search teams report regularly to the Incident Commander (IC) on their location progress and any victims found. A tactical channel (radio) is used.